Mingfeng HONG (洪铭锋)

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Research Interests: Open-source strategy; Open and distributed innovation; AI ecosystems and governance; digital platforms and online communities; AI-enabled innovation and strategy.

Methods: Computational social science; social network analysis; computational text analysis/NLP; configurational analysis; grounded theory; mixed-method research.

Education

University of Chinese Academy of Sciences Sep 2024 – Jul 2027 (expected)

- Hosted at Institutes of Science and Development, Chinese Academy of Sciences; M.S. Candidate in Management Science and Engineering.
- **GPA:** 3.95/4.00
- **Selected Awards:** National Scholarship; Outstanding Student Award; Best Paper Award.

Xidian University Sep 2020 – Jul 2024

- School of Economics and Management; Bachelor's Degree in Marketing.
- **GPA:** 3.7/4.00 (ranked 1st);
- **Selected Awards:** National Scholarship; Outstanding Graduate Award; Outstanding Undergraduate Thesis Award.

Selected Research Papers & Publications

- **Mingfeng Hong**, Zhisheng Hong*, Zixiang Wei
  From Stars to the Galaxy: Polycentric Governance, Digital Labor Division and Social Signaling as an Organizational Proxy in an Open-Source AI Ecosystem
  Revise & Resubmit (Major Revision): *Technological Forecasting and Social Change*. (JCR Q1, ABS 3)
  Presented at: 2025 China Technology Foresight Forum TFSC Paper Development Workshop. **Best Paper Award**.
  Research: Uses GitHub trace data from the LangChain ecosystem to examine how polycentric governance, contributor specialization and social signals organize distributed innovation in an open-source AI ecosystem.

- **Mingfeng Hong**, Zhisheng Hong*, Pengbo Liu
  Releasing the Crown Jewels? Unveiling the Paradox of LLM Open-Source Strategy via LLM-Assisted Grounded Theory
  Presented at: CNAIS 2025, Annual Conference of the Information Systems Engineering Committee, Systems Engineering Society of China.
  Research: Develops an LLM-assisted grounded theory from 1,361 public texts to explain how LLM firms strategically design openness to build ecosystems, shape standards, and capture value while managing platform dependence.

- Zhisheng Hong, **Mingfeng Hong**, Wenjiang Zheng*
  Antecedents of Firms' Ambidextrous Innovation Orientation under Human–AI Collaboration: A Configurational Analysis Based on the TOE Framework (in Chinese)
  Published: *Journal of Dongbei University of Finance and Economics (东北财经大学学报)*, 2025, (3), 14–29.
  Research: Examines how organizational and technological conditions jointly shape firms' ambidextrous innovation orientation under human–AI collaboration using configurational analysis based on the TOE framework.

- Zhisheng Hong, **Mingfeng Hong**
  Can Open-Source Collaboration Strengthen Digital Technology Ecosystem Resilience? Evidence from a Bilayer Network of Semiconductor Trade and Open-Source Collaboration (in Chinese)
  Presented at: The 22nd Annual Conference on China Science and Technology Policy and Management
  Research: Constructs a bilayer network linking semiconductor trade and GitHub collaboration across 149 economies to examine whether open collaboration provides resilience under partner disruptions, and how network structure and local capabilities condition that resilience.

- Zhisheng Hong, **Mingfeng Hong**
  Risk Identification and Runtime Governance of AI Agents from the Principal–Agent Perspective: An Analysis of the OpenClaw Open-Source Community (in Chinese)
  Presented at: The 22nd Annual Conference on China Science and Technology Policy and Management

- Zhisheng Hong, Yang Zhen, **Mingfeng Hong***
  Knowledge Management for Digital Enterprises (in Chinese)
  Published: *Tsinghua Business Review (清华管理评论)*, 2024, (11), 81–89

Policy Briefs: Contributed to Chinese Academy of Sciences policy research on emerging technologies and industrial transformation, including reports on China's Internet of Things development (2026) and the new round of scientific and technological revolution in agricultural industrial transformation (2025), endorsed with important written instructions.

Research Experience

Shanghai Artificial Intelligence Laboratory, Research Intern — Nov 2025 – Apr 2026

- Investigated open-source LLM ecosystems and the strategic design of model openness, focusing on how developer participation, incentives, and governance shape collaborative innovation and ecosystem development.
- Contributed to a research monograph on open-source innovation and co-developed a virtual research organization simulation system to compare alternative structures of scientific collaboration.
- Built computational research workflows combining large-scale platform trace data, text analysis, network analysis, and evidence synthesis for research on frontier AI and open innovation.

Selected Research Projects

National Social Science Fund of China, Research Team Member — 2025 – 2028 (expected)

- *User Innovation Enabled by Digital and Intelligent Technologies*.
- Contributed to the funded proposal and developed the research framework for collaboration between ordinary and lead users in online innovation communities, with particular attention to organizational boundary restructuring and public value creation.

National High-End Think Tank Key Project, Research Team Member — Apr 2025 – Nov 2025

- *Innovation Platform Networks and the Integration of Technological and Industrial Innovation*.
- Analyzed 33 National Manufacturing Innovation Centers, focusing on network structure, "company + alliance" governance, pilot-scale testing platforms, and cross-platform coordination; contributed to policy recommendations on innovation platform governance.

Ministry of Agriculture and Rural Affairs, Research Team Member — Jul 2024 – Mar 2025

- *Preliminary Research for China's 15th Five-Year Plan for Agricultural and Rural Development*.
- Assessed frontier technological and industrial trends in AI, synthetic biology, and new energy, and analyzed mechanisms through which emerging technologies diffuse into agricultural and rural sectors.

Open-Source Innovation & Leadership

PaperSeek, Creator and Core Maintainer — 2026 – present

- Created and lead the development of `paperseek/paperseek`, an open-source LLM-based scholarly-search agent integrating iterative retrieval, citation expansion, semantic retrieval/reranking, multi-provider routing, Web/CLI interfaces, and automated testing; **200+ GitHub Stars**.
- Lead system architecture, product roadmap, releases, documentation, and issue-driven iteration, translating user and community feedback into the development of the public research platform.

ModelScope Open-Source Community, Community Co-builder — 2026 – present

- Contribute to a large open-source AI ecosystem through research-oriented content, developer-facing knowledge sharing, and community initiatives around AI-assisted research and open-source tools; tutorials on research agents, BERT for social-science measurement, and PaperSeek were featured on ModelScope's homepage and official media channels.
- Co-organized the Beijing academic event "How to Work with Agents in Research" with ModelScope, Alibaba Cloud, UCAS, and Tsinghua student research communities.

Other Open-Source Work: Author of `css-research-skills` (30+ GitHub Stars), an open-source collection of AI-assisted social-science research workflows; contributor to `modelscope/Awesome-Vibe-Research` (400+ GitHub Stars).

Selected Coursework

Selected Coursework: Introduction to Social Computing; Text Data Analysis; Complex Systems Modeling and Simulation; Operations Research; Natural Language Processing and Deep Learning; Database Systems.

Methods & Technical Skills

- **Research Methods:** social network analysis; computational text analysis/NLP; statistical modeling; configurational analysis; Straussian grounded theory; qualitative/case research; mixed methods.
- **Programming & Data:** Python, R, SQL, Cypher, C, LaTeX; Git/GitHub, Docker, SQLite, APIs, graph databases.
- **Computational Research:** large-scale platform trace data collection and processing; network data analysis; simulation and system dynamics; LLM-assisted research workflows.